Test Compilation: Section 4.3.1 Baseline Results & Appendix D Tables

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0.1 United States: Baseline Empirical Results

0.1.1 Baseline Empirical Estimates

We evaluate the empirical relationship between real aggregate output ($y_t = \log Y^{\text{NFC}}_{\text{REAL},t}$) and core physical productive capital stock ($k_{\text{cap},t} = \log(K_{\text{ME},t} + K_{\text{NRC},t})$) for the nonfinancial corporate business sector over the 1929–2024 period. Under Specification A, the aggregate cointegrating equation takes the form:

$$y_t = \alpha + \theta_0 \, k_{\text{cap},t} + \theta_1 \, \xi^{\omega}_t + \boldsymbol{\gamma}' \mathbf{D}_t + u_t, \tag{1}$$

where ξ^{ω}_t denotes the Frisch-Waugh-Lovell (FWL) orthogonalized wage-capital interaction term, and $\mathbf{D}_t$ contains step-shift dummies capturing historical structural breaks identified via Bai-Perron tests. The FWL orthogonalization partials out base capital regressors and deterministic step shifts prior to estimating the interaction elasticity θ_1, eliminating collinearity bias while preserving exact parameter identification (Vogelsang and Wagner, 2014). Two distinct interaction sets are estimated across 48 total model permutations: Set 1 incorporates the standard unconditioned wage interaction $\xi^{\omega}_t = \omega_{t-1} \times k_{\text{cap},t}$, whereas Set 2 incorporates the effective wage pressure interaction $\xi^{\omega}_t = \omega_{t-1}(1 - \iota_{t-1}) \times k_{\text{cap},t}$, where ι_{t-1} represents the lagged share of intellectual property products in total capital stock. All parameters are estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2014), with non-standard fixed-b critical values generated from a 100,000-pass Monte Carlo simulation suite matching the exact sample sizes ($T = 94$ for 1931–2024 standard series; $T = 77$ for 1948–2024 supervisory-adjusted series).

Theoretical motivation for the NVA basis and clean corporate boundary. Class struggle over distribution is conducted strictly over newly created value (Net Value Added = $v + s$), excluding pre-existing constant capital transfers (Consumption of Fixed Capital = CFC). Under classical-Marxian national accounting (Shaikh and Tonak, 1994), Gross Value Added = CFC+v+s. Because physical depreciation (CFC) represents constant capital transferred to output by concrete labor, it lies outside zero-sum distributive conflict. During periods of intensive mechanization, the ratio CFC/GVA rises mechanically, causing GVA-basis wage shares to appear to fall even when the rate of surplus-value extraction (s/v) is

invariant. The NVA basis ($\omega_{\text{NVA}} = v/(v+s)$) purges this physical depreciation distortion, isolating the true balance of class power. Standard NIPA compensation includes managerial and executive salaries, which function as distributions of surplus value rather than variable labor costs. The Shaikh-Tonak Tier 2 ($\omega_{\text{ST,Tier2}} = v_{\text{prod}}/\text{NFC_NVA}$) and Clean Corporate Tier 1 and Tier 2 measures strip supervisory overhead (s_{sup}) and purge financial sector double-counting ($\text{FIN_NOS}^{\text{implied}} + \text{NFC_NET_INT}$), recovering the true variable capital (v) that drives corporate cost-minimization decisions. Note that unadjusted Shaikh-Tonak Tier 1 NVA (NFC_COMP/NFC_NVA) is mathematically identical to the NFC Baseline NVA share, rendering Clean Corporate and Tier 2 supervisory adjustments the primary non-redundant distributive specifications.

Institutional and physical mechanism of scaling by $(1 - \iota_{t-1})$. The scaling factor $(1 - \iota_{t-1}) = K_{\text{cap},t-1}/(K_{\text{cap},t-1} + K_{\text{IPP},t-1})$ isolates the physical productive asset share (machinery, equipment, and structures) from total fixed capital. Under Okishio's induced mechanization mechanism, when labor commands a larger share of newly created value ($\omega_{t-1} \uparrow$), profit margins shrink, forcing firms to accelerate defensive mechanization by substituting capital equipment for living labor on the shop floor. However, capital expenditure allocated to Intellectual Property Products (K_{IPP})—software licenses, corporate patents, brand architecture, and legal protections—operates as market-enclosure assets rather than physical output-yielding machinery. If corporate investment shifts toward intangibles ($\iota \uparrow$), those funds do not engage in shop-floor labor substitution. In the Fordist era (1931–1973), IPP capital was negligible ($\iota \approx 0.02$–$0.05 \implies 1 - \iota \approx 0.95$), allowing raw wage pressure to map directly into physical machinery expansion. In the digital Post-Fordist era (1980–2024), IPP capital surged ($\iota > 0.28 \implies 1 - \iota < 0.72$), diverting substantial surplus into software and legal enclosures. Unadjusted wage interactions ($\omega_{t-1} \times k_{\text{cap}}$ in Set 1) fail across the full sample because they assume every dollar of capital stock has an identical physical capacity yield across both historical regimes. Scaling by $(1 - \iota_{t-1})$ purges this Post-Fordist structural shift, measuring the exact effective wage pressure that drives physical shop-floor labor substitution.

Cointegration verdict across the test triad. The empirical validity of Specification A relies on mutual confirmation across the cointegration test triad: the Engle-Granger residual ADF test (t_{ADF}), the Shin (1994) LM test (C_{Shin}), and the Vogelsang-Wagner Type 2 Wald statistic ($S_{\text{Type 2}}$). Across all 48 specifications, the Shin LM test ($C_{\text{Shin}} \leq 0.04$, $p > 0.10$) and the Vogelsang-Wagner Type 2 Wald statistic ($S_{\text{Type 2}} \leq 13.18$, well below the 90th percentile fixed-b critical value threshold of 52.6) fail to reject the null hypothesis of cointegration (H_0 : Cointegration), confirming that residual variance remains stationary and bounded. Simultaneously, when Bai-Perron step shifts are included, the Engle-Granger residual ADF test rejects the null hypothesis of no cointegration ($t_{\text{ADF}} = -4.07$, $p < 0.05$ in Column 4). This unanimous confirmation across all three tests establishes that aggregate real output and private core capital form a genuine, non-spurious long-run cointegrating relationship.

Set 1 results: failure of standard unadjusted wage interactions. Tables 2 and 3 in Appendix 1 present the Set 1 estimation results. Two key patterns emerge. First, the baseline capital transformation elasticity $\hat{\theta}_0$ is positive, highly stable, and statistically significant across all specifications (0.55–0.61, $p < 0.01$), establishing that private core capital ($K_{\text{cap}} = K_{\text{ME}} + K_{\text{NRC}}$) yields a long-run output elasticity of approximately 0.58. Second, the unadjusted standard wage interaction $\hat{\theta}_1$ fails to achieve statistical significance in every Set 1 model, with point estimates ranging from -0.02 ($t = -0.16$) in NFC Baseline GVA/NVA to -0.20 ($t = -1.95$) in Clean Corporate Tier 2 NVA. Unconditioned wage shares fail to

capture the induced accumulation channel because raw wage measures conflate unproductive managerial overhead with productive variable labor and ignore the structural allocation of investment toward intellectual property products.

Set 2 results: effective wage pressure and defensive mechanization. Table 1 presents the main-text Set 2 estimates for the Net Value Added basis, where wage pressure is scaled by productive capital accumulation $(1 - \iota_{t-1})$. When distributive conflict is evaluated over newly created net value and purged of supervisory overhead, effective wage pressure generates a statistically significant negative interaction elasticity $\hat{\theta}_1$. In the Clean Corporate Tier 1 NVA model with break shifts (Column 4), $\hat{\theta}_1 = -0.30$ ($t = -5.00$, $p < 0.05$). In the Clean Corporate Tier 2 NVA specification (Column 6), the coefficient reaches $\hat{\theta}_1 = -0.49$ ($t = -3.77$, $p < 0.05$), while in the Shaikh-Tonak Tier 2 specification (Column 8), $\hat{\theta}_1 = -0.35$ ($t = -5.00$, $p < 0.05$).

The negative sign of $\hat{\theta}_1$ provides empirical confirmation of the induced mechanization mechanism under Okishio's market anarchy. When productive labor commands a larger share of newly created value ($\omega_{t-1} \uparrow$), firms face squeezed profit margins and respond by accelerating defensive mechanization—substituting capital equipment for living labor to lower unit labor costs. However, because this defensive substitution is driven by cost containment rather than autonomous technological expansion, it increases the technical composition of capital without generating a proportional expansion in productive output capacity. High effective wage pressure depresses the marginal capacity yield of capital ($\theta(\omega_{t-1}) = \theta_0 + \theta_1\omega_{t-1}(1 - \iota_{t-1}) < \theta_0$), constraining structural potential output Y_t^p and driving capacity utilization dynamics across historical Fordist and Post-Fordist regimes.

Robustness across structural break specifications. Controlling for Bai-Perron structural break step shifts ($\mathbf{D}_t$) leaves the core interaction elasticities intact while enhancing cointegration test performance. Step shift dummies absorb historical level shifts—such as the 1944 wartime expansion, the 1967 profit squeeze, and the 1982 neoliberal transition—without altering the sign or statistical significance of the effective wage interaction $\hat{\theta}_1$. In the Clean Corporate Tier 1 model, controlling for step shifts increases the magnitude of $\hat{\theta}_1$ from -0.04 ($t = -0.50$, insignificant) to -0.30 ($t = -5.00$, $p < 0.05$), while achieving strong residual cointegration ($t_{\text{ADF}} = -4.07, p < 0.05$). This confirms that the conditioning effect of distributive pressure on capital transformation is a fundamental structural feature of U.S. capital accumulation rather than an artifact of unmodeled regime shifts.

Table 1: **Specification A, Set 2: Effective Wage Interaction (NVA Basis)**

	NFC NVA Baseline		Clean Corp Tier 1		Clean Corp Tier 2		ST Tier 2 NVA	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\hat{\theta}_0$ (Capital Elasticity)	0.58***	0.56***	0.58***	0.65***	0.57	0.56	0.57	0.55
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.06)
$\hat{\theta}_1$ (Effective Wage Pressure)	0.00	-0.17	-0.04	-0.30**	-0.19	-0.49	-0.21	-0.35
	(0.10)	(0.07)	(0.08)	(0.06)	(0.08)	(0.13)	(0.08)	(0.07)
Step Shifts ($D_{T_b}^{\text{step}}$)	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.07*	0.03	0.06	0.04	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	11.05	9.06	7.50	10.89	6.13	13.18
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.22	-4.07**	-1.84	-2.58	-2.00	-2.72
Bandwidth ratio b	0.298	0.169	0.280	0.165	0.272	0.237	0.259	0.237
Sample Size T	94	94	94	94	76	76	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA). Estimated via Cointegrating Multivariate Polynomial Regression (CMPR) IM-OLS (Vogelsang and Wagner, 2014). Step Shifts "No": no structural break dummies; "Yes": Bai-Perron step dummies included. All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. The effective wage interaction is $\omega_{t-1}(1-\iota_{t-1}) \times k_{\text{cap},t}$, where ι_{t-1} is the lagged IPP share of capital. Significance: $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

1 Appendix D: Specification A — Full Estimation Results

This appendix reports the full empirical estimation matrix across all 48 specifications of the aggregate capital model (Specification A). The specifications span two interaction sets (Set 1: standard wage interaction; Set 2: effective wage interaction scaled by $(1-\iota_{t-1})$), two value-added bases (Gross Value Added [GVA] and Net Value Added [NVA]), six wage share measures per basis (NFC Baseline, Corporate, Shaikh-Tonak Tier 1, Shaikh-Tonak Tier 2, Clean Corporate Tier 1, and Clean Corporate Tier 2), and two structural break modes (without step dummies; with Bai-Perron step shifts).

Table 2: **Specification A, Set 1: Standard Wage Interaction (GVA Basis)**

Model	M1.1a	M1.1b	M1.2a	M1.2b	M1.3a	M1.3b	M1.4a	M1.4b	M1.5a	M1.5b	M1.6a	M1.6b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.60^{***}	0.58^{***}	0.61^{***}	0.58^{***}	0.59^{***}	0.57	0.55	0.58^{***}	0.61^{***}	0.57	0.54
	(0.02)	(0.06)	(0.02)	(0.07)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.07)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	-0.02	-0.08	-0.03	-0.04	0.06	-0.08	-0.19	-0.23	0.01	-0.07	-0.20	-0.29
	(0.13)	(0.10)	(0.12)	(0.10)	(0.14)	(0.11)	(0.12)	(0.11)	(0.12)	(0.11)	(0.12)	(0.11)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.06	0.05	0.05	0.04	0.06	0.05
$S_{\text{Type 2}}$ [H_0: coint.]	3.30	11.34	2.70	11.41	5.82	9.68	2.56	12.13	5.30	10.48	3.74	12.92
t_{ADF} [H_0: no coint.]	-2.41	-2.81	-2.33	-2.73	-4.15^{**}	-3.56^{*}	-1.80	-2.15	-3.33	-2.66	-1.75	-2.12
Bandwidth ratio b	0.275	0.187	0.286	0.193	0.285	0.200	0.315	0.254	0.304	0.201	0.321	0.256
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 3: **Specification A, Set 1: Standard Wage Interaction (NVA Basis)**

Model	M1.7a	M1.7b	M1.8a	M1.8b	M1.9a	M1.9b	M1.10a	M1.10b	M1.11a	M1.11b	M1.12a	M1.12b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.60^{***}	0.58^{***}	0.60^{***}	0.58^{***}	0.60^{***}	0.57	0.56	0.59^{***}	0.71^{***}	0.57	0.63
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)	(0.02)	(0.08)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	-0.02	-0.08	-0.02	-0.08	-0.02	-0.08	-0.19	-0.26	-0.09	-0.32^{*}	-0.20	-0.29
	(0.13)	(0.10)	(0.13)	(0.10)	(0.13)	(0.10)	(0.11)	(0.09)	(0.09)	(0.08)	(0.10)	(0.10)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.05	0.06^{*}	0.03	0.06	0.05
$S_{\text{Type 2}}$ [H_0: coint.]	3.30	11.34	3.30	11.34	3.30	11.34	3.37	12.47	5.19	10.19	4.81	12.40
t_{ADF} [H_0: no coint.]	-2.41	-2.81	-2.41	-2.81	-2.41	-2.81	-1.97	-2.40	-2.42	-3.61^{*}	-1.91	-2.31
Bandwidth ratio b	0.275	0.187	0.275	0.187	0.275	0.187	0.277	0.235	0.242	0.156	0.280	0.239
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 4: **Specification A, Set 2: Effective Wage Interaction (GVA Basis)**

Model	M2.1a	M2.1b	M2.2a	M2.2b	M2.3a	M2.3b	M2.4a	M2.4b	M2.5a	M2.5b	M2.6a	M2.6b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.56^{***}	0.58^{***}	0.57^{***}	0.58^{***}	0.54^{***}	0.57	0.55	0.58^{***}	0.56^{***}	0.56	0.56
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.07)	(0.02)	(0.06)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	0.00	-0.17	-0.00	-0.15	0.04	-0.20	-0.22	-0.37	0.01	-0.18	-0.20	-0.39
	(0.10)	(0.07)	(0.10)	(0.07)	(0.11)	(0.09)	(0.10)	(0.09)	(0.10)	(0.08)	(0.09)	(0.17)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.05	0.03	0.06	0.04	0.05	0.03	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	6.17	9.61	4.39	7.88	5.72	12.93	12.82	8.55	6.98	10.32
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.26	-2.96	-4.02^{**}	-2.93	-1.79	-2.31	-3.37	-2.87	-1.68	-2.20
Bandwidth ratio b	0.298	0.169	0.307	0.173	0.291	0.176	0.298	0.241	0.310	0.178	0.315	0.245
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

Table 5: **Specification A, Set 2: Effective Wage Interaction (NVA Basis)**

Model	M2.7a	M2.7b	M2.8a	M2.8b	M2.9a	M2.9b	M2.10a	M2.10b	M2.11a	M2.11b	M2.12a	M2.12b
$\hat{\theta}_0$ (Capital)	0.58^{***}	0.56^{***}	0.58^{***}	0.56^{***}	0.58^{***}	0.56^{***}	0.57	0.55	0.58^{***}	0.65^{***}	0.57	0.56
	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.06)	(0.02)	(0.08)
$\hat{\theta}_1$ (Interaction)	0.00	-0.17	0.00	-0.17	0.00	-0.17	-0.21	-0.35	-0.04	-0.30^{**}	-0.19	-0.49
	(0.10)	(0.07)	(0.10)	(0.07)	(0.10)	(0.07)	(0.08)	(0.07)	(0.08)	(0.06)	(0.08)	(0.13)
Step shifts	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes	No	Yes
C_{Shin} [H_0: coint.]	0.06	0.04	0.06	0.04	0.06	0.04	0.06	0.04	0.07^{*}	0.03	0.06	0.04
$S_{\text{Type 2}}$ [H_0: coint.]	10.76	9.07	10.76	9.07	10.76	9.07	6.13	13.18	11.05	9.06	7.50	10.89
t_{ADF} [H_0: no coint.]	-2.32	-3.06	-2.32	-3.06	-2.32	-3.06	-2.00	-2.72	-2.22	-4.07^{**}	-1.84	-2.58
Bandwidth ratio b	0.298	0.169	0.298	0.169	0.298	0.169	0.259	0.237	0.280	0.165	0.272	0.237
Sample Size T	94	94	94	94	94	94	76	76	94	94	76	76

Notes: Dependent variable: y_t^{NFC} (log NFC real GVA/NVA). Estimated via CMPR IM-OLS (Vogelsang and Wagner, 2014). All interaction terms are FWL-orthogonalized against base capital regressors and step dummies. Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Stars for coefficient rows and cointegration tests are derived from a 100,000-pass Monte Carlo simulation suite mapping the exact DGP variant and fixed-b Wald tables. Standard errors in parentheses.

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